

$$G \times X \rightarrow X$$

证明: 类似数集.

$$G: \{A \in M_n(\mathbb{R}) \mid A^T A = E\} \text{ 称为正交变换群}$$

$$X = \mathbb{R}^n$$

$$G \times X \rightarrow X$$

$$(A, \alpha) \mapsto A\alpha.$$

$$\alpha = 0 \cdot G\alpha = \{0\}.$$

$$\forall \alpha \in \mathbb{R}^n \rightarrow \alpha \neq 0 \quad G\alpha = \{\beta \mid |\beta| = |\alpha|\}.$$

$$A\alpha, \beta \rightarrow (A\alpha, A\beta) = (A\alpha)^T A\beta = \alpha^T \beta.$$

$$\text{ii) } |\alpha| = |\beta| \neq 0.$$

证毕 2.1.7

$$\mathbb{R}^n = \text{Span}\{B_i\}$$

$$\alpha, \beta \in B \quad \alpha \neq \beta \quad \mathbb{R}\alpha \cap \mathbb{R}\beta = \{0\}$$

$$\mathbb{R}^* \times V \rightarrow V.$$

Prop 1.3.2.  $X$  is a  $G$ -set.

$$\forall x, y \in X.$$

$$G \cdot x \cap G \cdot y = \begin{cases} \emptyset \\ Gx = Gy. \end{cases}$$

$$\textcircled{*} X = \bigcup_{x \in B} Gx \quad \text{这组的作用是找“基”}$$

佐思引理: (Zorn lemma)

pf: If  $Gx \cap Gy \neq \emptyset$

$$\exists g_1, g_2 \in G \text{ s.t. } g_1x = g_2y$$

$$\Rightarrow x = g_1^{-1}g_2y = (g_1^{-1}g_2)y \in G \cdot y$$

$$\Rightarrow \forall h \in G \quad hx = (hg_1^{-1}g_2)y \in Gy \Rightarrow Gx \subseteq Gy$$

Similarly,  $Gy \subseteq Gx \Rightarrow Gx = Gy$

$$B = \{B_i \in X \mid \bigcup_{x \in B_i} Gx, \forall x \neq y \in B_i \quad Gx \cap Gy = \emptyset\}$$

$$= \bigcup_{x \in B_i} Gx \text{ 互不交.}$$

$$(i) \emptyset \neq B_i \quad |B_i| = 1. \quad \bigcup_{x \in B_i} Gx = \bigcup_{x \in B} Gx.$$

$$(i) B_1 \subseteq B_2 \quad B_i \in B \Leftrightarrow B_i \subseteq B_2. \text{ (partial order. set.)}$$

$$(ii) \text{ suppose } B_i \subseteq B_{i+1} (i=1, \dots) \Rightarrow B \in B$$

$$\text{s.t. } B \subseteq B$$

Then there is a maximal element  $B^*$  in  $G$ .

$$(i) B \subseteq B^* \Rightarrow B \subseteq B^* \forall B$$

Proof of (iii)

$$\text{Let } B = \bigcup B_i.$$

$$\Rightarrow B \in B. \forall x, y \in B$$

$$Gx \cap Gy = \emptyset$$

$$\text{Since } x, y \in B = \bigcup B_i$$

$$\therefore \exists n. \text{ s.t. } x, y \in B_n \Rightarrow Gx \cap Gy \neq \emptyset \text{ (矛盾)}$$

且  $B$  是一个极大.

$$\text{Then } X = \bigcup_{x \in B} Gx.$$

$$\text{要证:}$$

$$\exists x, y \in X \setminus (\bigcup_{x \in B} Gx)$$

$$\Rightarrow x \notin \bigcup_{x \in B} Gx \Rightarrow Gx \cap (\bigcup_{x \in B} Gx) = \emptyset$$

$$= \bigcup_{x \in B} (Gx \cap Gx)$$

By (i) (if  $Gx \cap Gy \neq \emptyset$  then  $Gx = Gy$ )

$$\Rightarrow \exists x_0 \in \bigcup_{x \in B} Gx. \text{ (矛盾)}$$

$$\Rightarrow Gx_0 \cup (\bigcup_{x \in B} Gx) = \bigcup_{x \in B \cup \{x_0\}} Gx$$

$$X = M_{\min}(\mathbb{R}).$$

$$G = GL(m, \mathbb{R}) \times GL(n, \mathbb{R})$$

$$= \{ (A, B) \mid A \in GL(m, \mathbb{R}) \mid B \in GL(n, \mathbb{R}) \}$$

$$(A, B)(C, D) = (AC, BD)$$

$$G \times X$$

$$(A, B), M \mapsto AXB^T \text{ (或 } B^{-1})$$

$$X \text{ and } AXB^T$$

$$B = \left\{ \begin{pmatrix} E_r & 0 \\ 0 & 0 \end{pmatrix} \mid 0 \leq r \leq \min\{m, n\} \right\}$$

$$\mathbb{R} \times V \rightarrow V. \alpha \neq 0 \quad \mathbb{R} \xrightarrow{T, T} \mathbb{R}\alpha$$

$$\mathbb{R}^* \times V \rightarrow V$$

Prop 1.3.5  $X$  is a  $G$ -set.  $\forall x \in X$

$$|Gx| |G \cdot x| = |G|$$

$$|A \times B| := |A \times B|$$

$$|G \times G| = |G|$$

$$\{ (h, g) \mid h \in G, g \in G \} \xrightarrow{1:1} G$$

$$G \cdot x = \{ gx \mid g \in B \} \text{ if } g \neq g_2 \in B \text{ } g_2 \neq g_1$$

注:  $G_x$  是  $x$  的稳定子群.  $G \cdot x$  是  $x$  的轨道.

$$G_x \times G_x \xrightarrow{1:1} G_x \times B$$

$$\text{定义: } \varphi: G_x \rightarrow G_x$$

$$|G \cdot x| = \frac{|G|}{|Ker \varphi|} = \frac{|G|}{|G_x|}$$

$$\Rightarrow |G| = |G \cdot x| |G_x|$$

即: 第一同构定理与陪集 (Card 相等)

$$\varphi: (a, b) \mapsto ab$$

$$ha = ba \Rightarrow a_2 a_1 = b_2^{-1} b_1$$

$$b_2^{-1} b_1 x = x \Rightarrow b_2 x = b_1 x$$

由  $B$  的选取,  $b_1 = b_2$

( $B \triangleright G$  是单射)

$$a_2^{-1} a_1 = b_2^{-1} b_1 e \Rightarrow a_1 = a_2 \therefore \text{这是单射}$$

$$\forall g \in G \quad g x \in \{ h x \mid h \in B \} = G \cdot x$$

$$\Leftrightarrow g x = h x \text{ for some } h \in B$$

$$\Rightarrow h^{-1} g \cdot x = x \Rightarrow h^{-1} g \in G_x$$

$$\Rightarrow \varphi(h, h^{-1}g) = g \therefore \text{这是一个映射}$$

$$\text{Example: } H = G, (H \cdot x) \times G \rightarrow G$$

$$(a, b), g \mapsto agb$$

$$(H \cdot x)g = \{ agb \mid a \in H, b \in L \} = H g L$$

叫做双陪集 ( $\alpha$  bicosa)

$$\text{即: } G = \bigcup_{g \in B} H g L$$

另一个例子: 取  $G = GL(n, \mathbb{R})$

$$H = B(n, \mathbb{R}) = \left\{ \begin{pmatrix} a_1 & * \\ & \ddots \\ 0 & a_n \end{pmatrix} \mid a_1, \dots, a_n \neq 0 \right\} \text{ 即: 上三角可逆群}$$

$$L = B(n, \mathbb{R})^T$$

$$GL(n, \mathbb{R}) = \bigcup_{w \in W} B(n, \mathbb{R}) w B(n, \mathbb{R})^T$$

$$L = \{ e \}$$

$$H g L := H g$$

left coset 左陪集

$H = \{ e \}$  右陪集

$$G = \bigcup_{a \in B} H a = \bigcup_{a \in B} a H$$

$$\text{Prop } |B| = |B|$$

$$:= [G:H]$$

index of  $H$  in  $G$

$$B_1' = \{ H a \mid a \in B_1 \} \quad B_2' = \{ a H \mid a \in B_2 \}$$

$$|B_1'| = |B_1| \quad |B_2'| = |B_2|$$

$$\text{Prop: } |B_1'| = |B_2'|$$

$$\varphi: H a \mapsto a^{-1} H$$

$$H a_1 = H a_2 \Rightarrow a_1 a_2^{-1} \in H \Rightarrow (a_1^{-1})^{-1} a_2^{-1} \in H$$

$$(a_1^{-1})^{-1} a_2^{-1} H = H \Rightarrow a_2^{-1} H = a_1^{-1} H$$

$$\varphi^{-1}(a H) = H a^{-1}$$

$$|G| = \sum_{a \in B_1} |H a| = |B_1| |H| \Leftrightarrow |G| = |H| [G:H]$$

$$\begin{matrix} |H| & (H \rightarrow H a) \\ x \mapsto x a \\ (y a) a^{-1} \mapsto y a \end{matrix}$$

$$\text{Cor. } |H| \mid |G|$$

Suppose  $G$  is a group and  $|G| = p$  where  $p$  is prime. Then

where  $p$  is prime. Then

$$G = \langle a \rangle \quad |a| = p$$

$$\text{proof } \forall a \in G \quad a \neq e$$

$$\langle a \rangle \subseteq G \quad |\langle a \rangle| > 1$$

$$|\langle a \rangle| \mid |G|$$

$$|\langle a \rangle| = p$$

$$\langle a \rangle \subseteq G \Rightarrow G = \langle a \rangle$$

Prop 1.3.

$$\frac{H}{K} = G \quad H/K := \{ ab \mid a \in H, b \in K \}$$

$$1) |H/K| = |K| [H : H \cap K]$$

$$= |H| [K : H \cap K]$$

$$(2) \Rightarrow K = G$$

$$(3) |HaK| = \frac{|H||K|}{|H \cap Ka|}$$

$$\text{proof: Let } X = \{ aK \mid a \in H \}$$

$$|HK| = |X| |K| \text{ and } |K| = |K|$$

$$\text{Def II: } X \xleftrightarrow{1:1} \{ a(H \cap K) \mid a \in H \}$$

$$H \cap K \leq H$$

$$aK \mapsto a(H \cap K) \text{ surjective}$$

$$a_1 K = a_2 K \Rightarrow a_1^{-1} a_2 \in K$$

$$a_i \in H \text{ thus } a_1^{-1} a_2 \in H \cap K$$

$$\Rightarrow a_1(H \cap K) = a_2(H \cap K)$$

$$a_1(H \cap K) = a_2(H \cap K)$$

$$a_1^{-1} a_2 \in H \cap K \Rightarrow a_1^{-1} a_2 \in K$$

$$\therefore a_1 K = a_2 K$$

$$\therefore aK = bK$$

injective

$$|HaK| = |H/Ka| = |H(aKa^{-1})|$$

$$= \frac{|H| |aKa^{-1}|}{|H \cap aKa^{-1}|} = \frac{|H||K|}{|H \cap Ka|}$$

$$K \longrightarrow aKa^{-1}$$

$$x \longrightarrow axa^{-1}$$

$$a^{-1}ya \longleftarrow y$$

$$H \leq G$$

$$G/H := \{ aH \mid a \in G \}$$

$$H \backslash G = \{ Ha \mid a \in G \}$$

$$H \backslash G / K = \{ HaK \mid a \in G \}$$

$$S_3 = \{ (1), (12), (13), (23), (123), (132) \}$$

$$H = \{ (1), (12) \}$$